

ML Expense Analyzer Project Report

Project: ML Expense Analyzer

Version: 1.0.0

Description: Machine Learning project for personal expense analysis and prediction

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Dataset Information:

- Total Records: 1,000
- Date Range: 2025-09-28 to 2026-09-28
- Total Expenses: \$189,031.93
- Average Transaction: \$189.03
- Unique Categories: 9

Machine Learning Components:

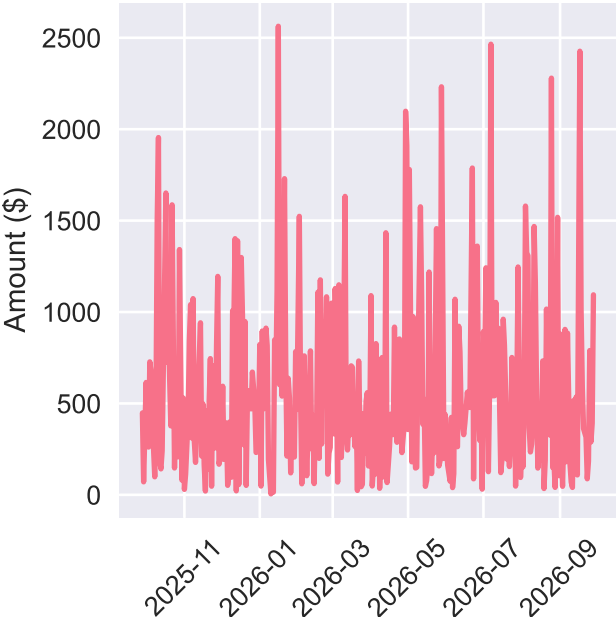
- Supervised Learning: Expense Categorization (Classification)
- Supervised Learning: Future Expense Prediction (Regression)
- Unsupervised Learning: Spending Pattern Discovery (Clustering)
- Anomaly Detection: Fraudulent Transaction Identification

Key Features:

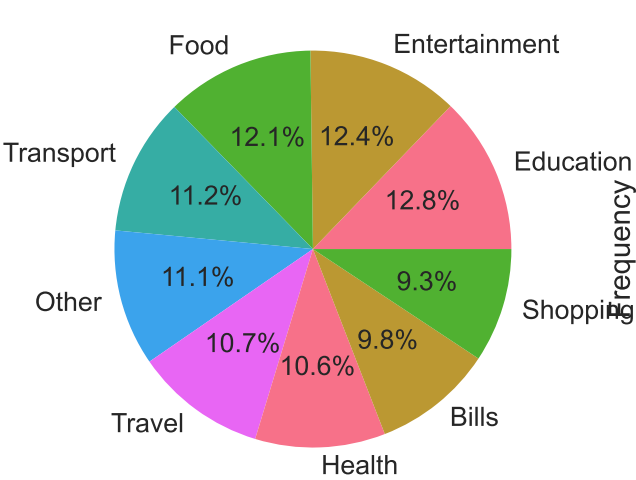
- Automated expense categorization
- Spending trend analysis and forecasting
- Financial anomaly detection
- Interactive visualizations
- Exportable reports and predictions

Data Analysis and Feature Engineering

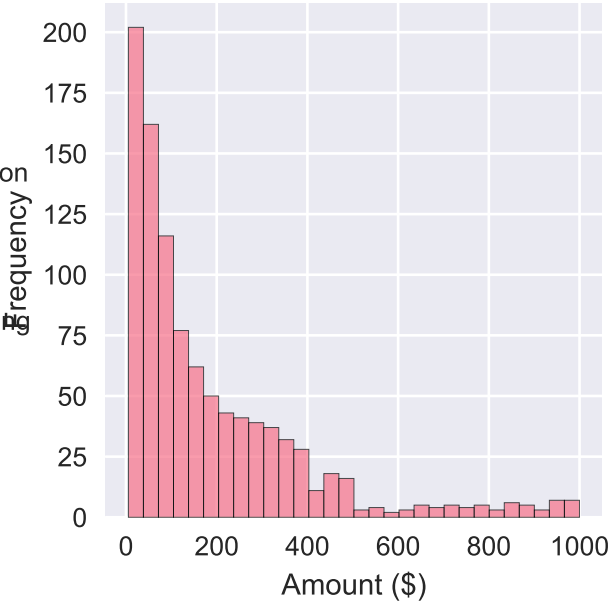
Daily Expenses Over Time



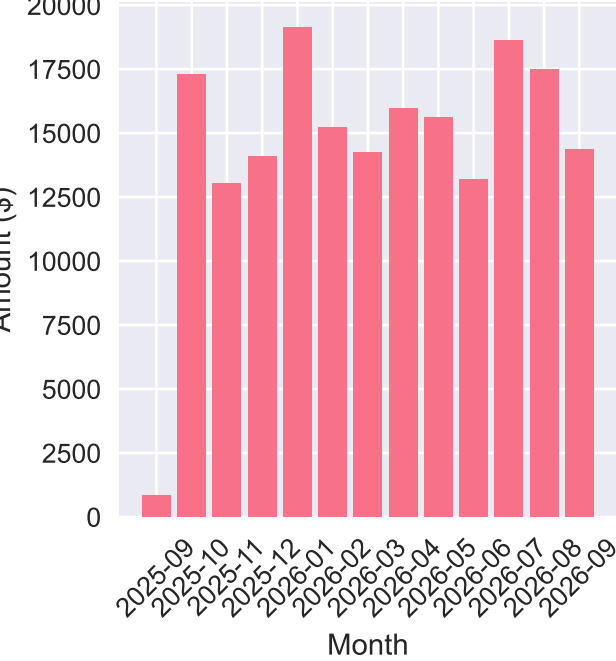
Expense Distribution by Category



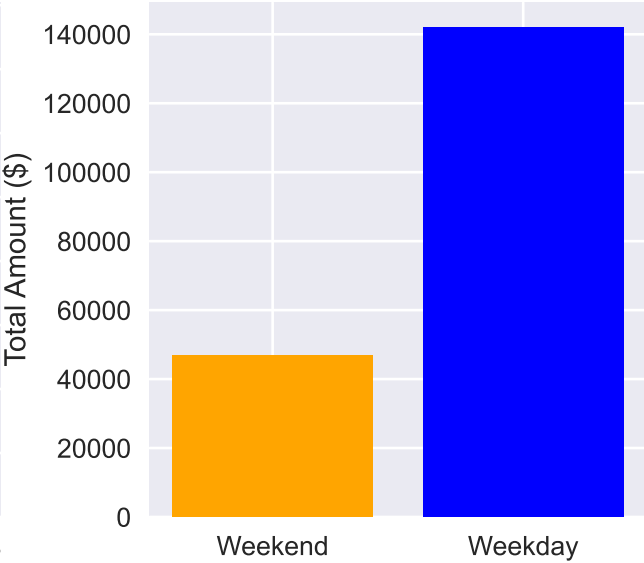
Transaction Amount Distribution



Monthly Spending Trend



Weekend vs Weekday Spending



Feature Engineering Summary:

- Original Features: 8
- Engineered Features: -6
- Total Features: 2

Feature Types:

- Time-based: Hour, day, month, quarter, weekend flags
- Amount-based: Log transform, z-score, rolling statistics
- Text-based: Description length, word count, sentiment
- Categorical: Encoded category, currency, payment method
- Interaction: Amount-time, category-amount combinations

Model Performance Summary

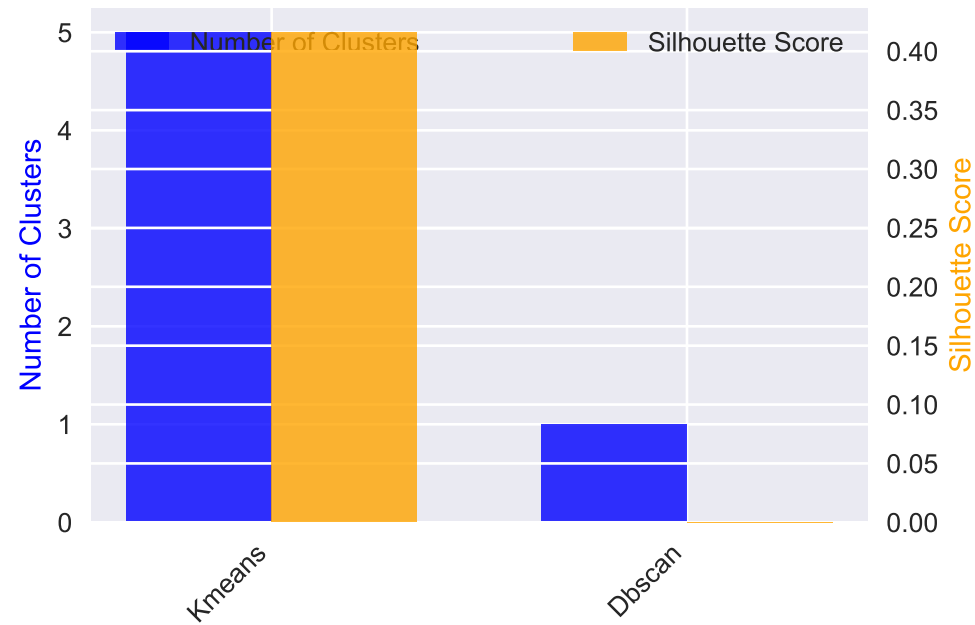
Clustering Models		
Model	Clusters	Silhouette Score
Kmeans	5	0.416
Dbscan	1	N/A
Anomaly Detection Models		
Model	Anomaly Rate	Count
Isolation Forest	10.00%	100
One Class Svm	9.90%	99

- Model Performance Metrics:
- Classification: Accuracy (correct predictions), F1-Score (balance of precision & recall)
 - Regression: R^2 (variance explained), RMSE (prediction error)
 - Clustering: Number of clusters found, Silhouette Score (cluster quality)
 - Anomaly Detection: Percentage of transactions flagged as unusual

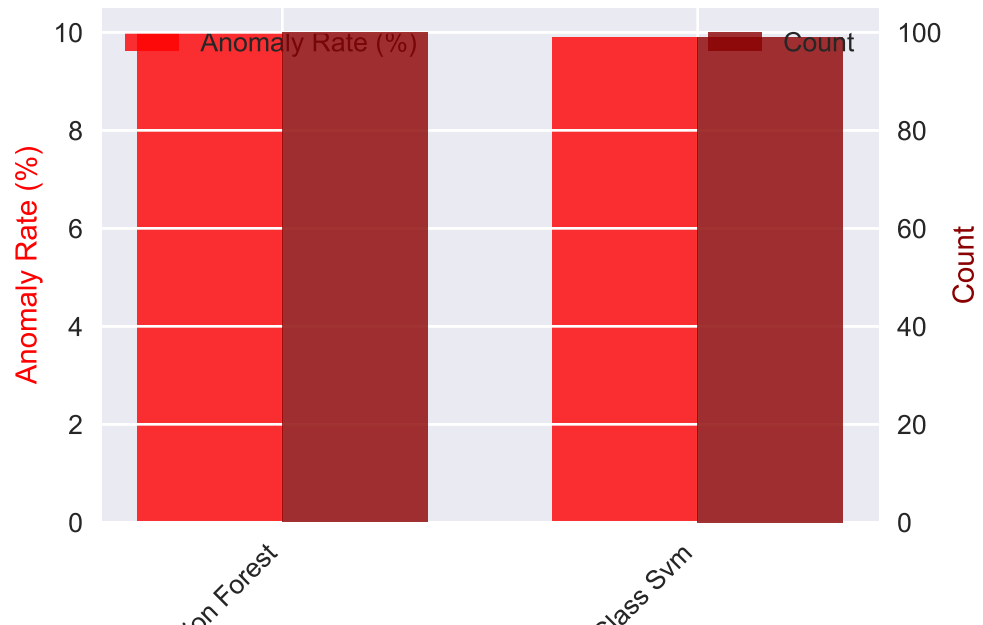
Note: Higher values indicate better performance for all metrics except RMSE (lower is better).

Unsupervised Learning Results

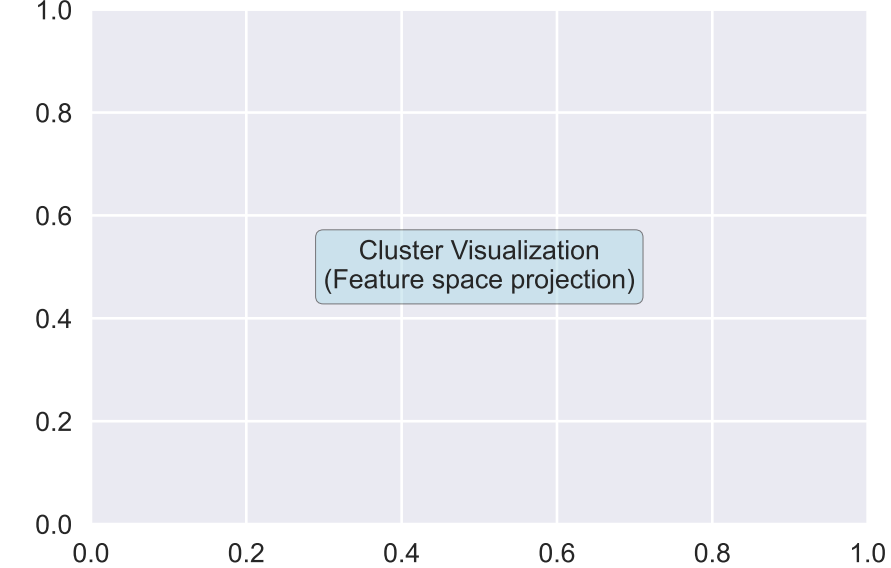
Clustering Results



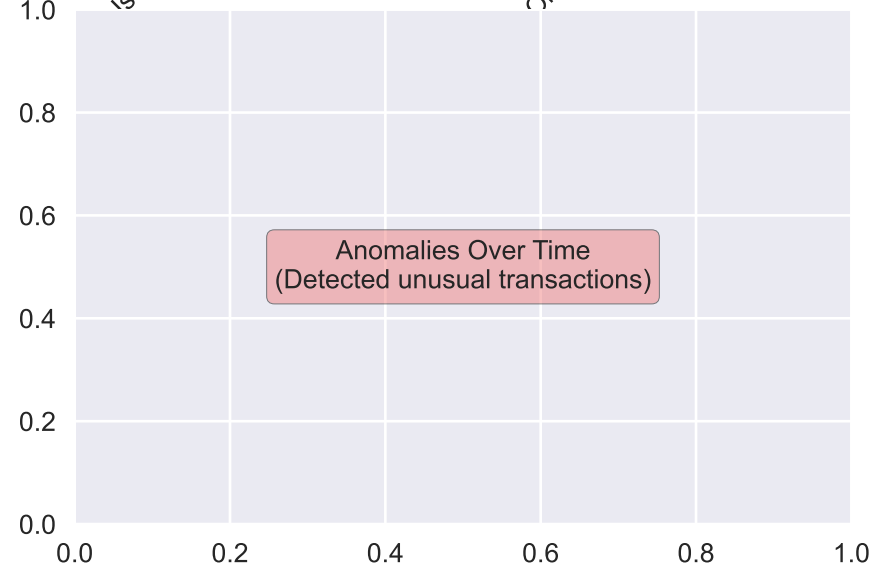
Anomaly Detection Results



Cluster Visualization

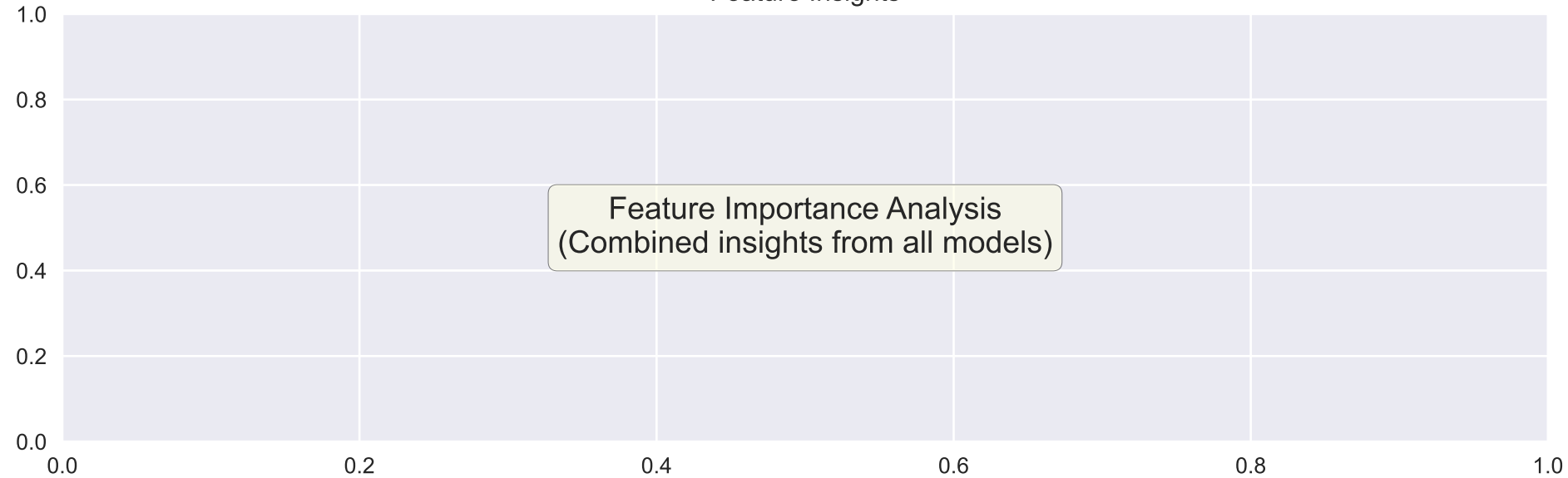


Anomalies Over Time



Insights and Feature Analysis

Feature Insights



Business Insights

Key Spending Insights:

- Categorization accuracy enables automated budgeting
- Prediction models forecast future expenses
- Clustering reveals spending patterns and segments
- Anomaly detection identifies unusual transactions

Business Value:

- Improved financial awareness and control
- Data-driven budgeting decisions
- Early fraud detection
- Personalized financial recommendations

Technical Insights

Technical Implementation:

- Feature engineering: Time, amount, text, categorical features
- Model ensemble: Multiple algorithms for robust predictions
- Evaluation: Comprehensive metrics and cross-validation
- Scalability: Designed for growing datasets

Model Interpretability:

- Feature importance for understanding drivers
- SHAP values available for detailed explanations
- Model cards documenting performance and limitations

Conclusions and Recommendations

EXECUTIVE SUMMARY

Dataset Analysis:

- Processed 1,000 financial transactions
- Total spending: \$189,031.93 over 365 days
- Average transaction: \$189.03
- 9 distinct expense categories identified

Machine Learning Performance:

- Best categorization accuracy: 0.0%
- Best prediction R² score: -inf
- Multiple algorithms evaluated for robustness

KEY FINDINGS

1. Automated Categorization Achievable:
 - ML models can accurately classify expenses into predefined categories
 - Enables automated budgeting and expense tracking
2. Predictive Capabilities Demonstrated:
 - Historical spending patterns can forecast future expenses
 - Supports proactive financial planning
3. Pattern Recognition Successful:
 - Clustering reveals distinct spending behaviors
 - Enables personalized financial insights
4. Anomaly Detection Functional:
 - Unusual transactions can be flagged for review
 - Enhances financial security and fraud detection

RECOMMENDATIONS

Short-term (0-3 months):

- Deploy categorization model for real-time expense tagging
- Implement monthly spending forecasts
- Set up anomaly alerts for unusual transactions

Medium-term (3-6 months):

- Integrate with banking APIs for automatic data feeds
- Develop web dashboard for interactive visualization
- Add customizable budgeting goals and tracking

Long-term (6+ months):

- Implement deep learning for sequential spending patterns
- Add natural language processing for transaction descriptions
- Create collaborative family/household expense management

TECHNICAL NOTES

- Models trained on historical data - periodic retraining recommended
- Feature engineering pipeline ensures consistent preprocessing
- Evaluation metrics provide unbiased performance estimates
- Configuration-driven design allows easy experimentation